UNIT V

Building Fire Safety

fire-safe design objectives | fire load | fire resistant materials & testing | structural protection | egress & exit width | fire certificates | high-rise

I. Objectives of fire-safe building design

Fire safety is built into a building at the **design stage**: layout, materials, structure, services and escape.

Reference code in India: **National Building Code of India (NBC 2016), Part 4 – Fire and Life Safety**, with IS codes and state fire rules.

Objective	How it is achieved
Life safety	Safe, quick escape for all occupants incl. disabled; protection of fire-fighters; smoke control.
Property protection	Limit spread to the room / compartment of origin; non-combustible construction; sprinklers.
Structural stability	Building does not collapse for the rated time so that people can leave and fire fighters can work.
Business continuity	Protect critical processes, records; quick recovery; insurance requirements.
Environment protection	Control fire water run-off & toxic smoke; avoid damage to neighbours.
Legal compliance	NBC, Fire Act/Rules of the State, Fire NOC, insurance conditions.

Basic strategy



> **Fire safety triad**: compartmentation, means of escape, fire fighting access - each independent, so that if one fails others work.

> **Occupancy classification** (NBC): Residential (A), Educational (B), Institutional (C), Assembly (D), Business (E), Mercantile (F), Industrial (G), Storage (H), Hazardous (J).

> **Types of construction** (NBC): Type I (highest fire resistance, 4 hr) to Type 4 (lowest) - depends on height, area and occupancy.

II. Fire load

Fire load is the **total amount of heat energy** that can be released by complete burning of all combustible contents of a building / compartment, including finishes, structure and furniture.

$$\text{Fire load } q = (\text{Sum of } M \times H) / A \quad [MJ/m2]$$

> **M** = mass of each combustible (kg), **H** = its calorific value (MJ/kg), **A** = floor area of compartment (m2).

> Often expressed as **equivalent kg of wood per m2**; calorific value of wood is taken as about 18-19 MJ/kg (so 1 kg/m2 wood = about 18.8 MJ/m2).

> **Fire load density** (per m2 of floor area) decides the severity & duration of a fire and so the **fire resistance** needed for the structure.

Fire load category (NBC)	Typical occupancy	Remark
Low	Dwellings, offices, hospitals, schools, hotels	Mostly furniture & paper; < about 1.1 GJ/m2 (1100 MJ/m2)
Moderate	Workshops, shops, factories, libraries, storage of non-hazardous items	About 1.1 - 2.2 GJ/m2
High	Warehouses of plastics/rubber/paper, timber, chemicals, hazardous storage	Above 2.2 GJ/m2 - need highest protection

Worked example

A store of 5 m x 4 m (A = 20 m2) holds 400 kg of wood (H = 18.8 MJ/kg), 100 kg of paper (H = 17 MJ/kg) and 50 kg of plastic (H = 40 MJ/kg). Total = 400x18.8 + 100x17 + 50x40 = 7520 + 1700 + 2000 = 11,220 MJ. Fire load = 11,220 / 20 = **561 MJ/m2** (= about 30 kg/m2 of wood) -> **low category**.

Fire severity & duration

> Equivalent fire duration (hrs) is about **fire load (kg wood/m2) / 40-50 x ventilation factor**; e.g. 30 kg/m2 -> about 0.75 hr burn at standard severity.

> More ventilation -> faster, hotter fire; less ventilation -> slower, longer fire (ventilation-controlled).

> Fire growth stages: **incipient -> growth -> flashover -> fully developed -> decay**. Flashover (about 500-600 C) is the point after which survival is unlikely.

Stage	What happens	Action
Incipient	Smouldering, little heat	Detection, extinguish with extinguisher
Growth	Flames spread, smoke layer, temperature rise	Escape, sprinklers act
Flashover	All surfaces ignite suddenly	Late - structure must hold
Fully developed	Maximum heat release	Compartment walls & structure resist
Decay	Fuel consumed; cooling	Danger of backdraught, collapse

III. Fire resistant materials

Fire resistance is the ability of a building element (wall, floor, beam, door) to **retain stability, integrity and insulation** for a stated time when exposed to fire. Rated in hours (1/2, 1, 1.5, 2, 3, 4 hr). A **non-combustible** material neither burns nor adds to fire.

Material	Behaviour in fire / use
Brick & masonry	Non-combustible; 230 mm brick wall about 4 hr; bad conductor.
Reinforced concrete (RCC)	Good: concrete cover protects steel; 2-4 hr depending on size & cover. Risk: spalling of concrete at high temperature.
Structural steel	Non-combustible but loses 50% strength at about 550-600 C -> needs protection.
Gypsum plaster / boards	Contains 21% bound water; releases as steam, keeps the temperature at about 100 C; used for ceilings, partitions.
Vermiculite / perlite / mineral-wool sprays	Light insulating coatings on steel.
Intumescent paints & sealants	Swell to a thick insulating char at 200-250 C; thin, decorative.
Fire doors & glass	Steel / timber core doors 1-2 hr rated; wired, laminated or ceramic fire glass.
Timber (treated)	Combustible but large sections char slowly (0.6-0.8 mm/min); fire-retardant treatment.
Aluminium & plastics	Aluminium melts at 660 C; plastic/PVC give smoke, toxic gases - avoid in escape routes.

Classification of materials (NBC / IS)

> **Non-combustibility test** (IS 3808 / ISO 1182): furnace test at 750 C; material is non-combustible if no flaming and a temperature rise less than 50 C.

> **Surface spread of flame** (IS 12777 / BS 476 Part 7): Class 1 (best) to Class 4 - for wall & ceiling linings.

> **Fire propagation index / ignitability, smoke density, toxicity** also considered for finishes, curtains and carpets in exits.

IV. Fire testing

Fire resistance is **proved by standard test**: a full-size element is exposed to the **standard time-temperature curve** in a furnace while loaded.

ISO 834 curve: $T = 345 \log10 (8t + 1) + 20$ (T in C, t in minutes)

Time (min)	Furnace temp (C)	Time (min)	Furnace temp (C)
5	576	60	945
15	739	120	1049
30	842	240	1153

Fire resistance - 3 failure criteria

Criterion	Meaning	Failure when
Stability (R)	Load-bearing capacity	Collapse or excessive deflection
Integrity (E)	Stops passage of flame / hot gases	Cracks, openings; cotton pad ignites
Insulation (I)	Restricts heat transmission	Average rise on unexposed face > 140 C or any point > 180 C

> Test standards: **IS 3809 (Fire resistance test of structures), ISO 834, BS 476 Part 20-22, ASTM E119, UL 263.**

> A rating '**60/60/60**' = 60 min for stability / integrity / insulation. Doors tested for **FD30 / FD60** etc. Other tests: smoke density, flame spread, fire-door and sprinkler component tests.

> **Large-scale and fire engineering**: computer models (CFD, zone models) are used for unusual buildings, with real fire scenario / design fire.

V. Structural fire protection

Methods to keep the temperature of structural members below the **critical temperature** (steel about 550 C; concrete reinforcement about 400-500 C) for the rated time.

Method	Description
Concrete encasement / cover	Steel columns or beams cast in concrete; reinforcement given 25-75 mm cover.
Boards / cladding	Gypsum, calcium silicate or vermiculite boards fixed around steel; hung ceilings and fire-rated partitions.
Spray-applied materials	Cementitious or mineral-fibre sprays on steel; fast, cheap, rough finish.
Intumescent coating	Paint expands into insulating foam char; used where appearance matters; 30-120 min.
Water-filled / concrete-filled hollow columns	Heat is absorbed by water or concrete core; used in some tall buildings.
Compartmentation	Divide building into fire compartments with fire-rated walls and floors; limits area & spread (NBC: horizontal & vertical).
Fire stopping & dampers	Seal service penetrations, ducts, shafts, cable trays.
Separation distance	Space between buildings & external wall construction to prevent spread by radiation.

Typical required fire resistance (NBC, hours)

Element	Type 1 (high-rise)	Type 2	Type 3
Load-bearing walls & columns	4	3	2
Beams	3	2	2
Floors	2	2	1.5
Non-load-bearing partitions	1-2	1	1

Note

> Values are indicative; always check the NBC 2016 Part 4 tables for the occupancy, height and area when designing.

VI. Structural integrity in fire

> Structural integrity = ability of the structure to **remain standing and avoid disproportionate (progressive) collapse** during and after fire.

> **Steel**: yield strength falls to about 50% at 550-600 C; thermal expansion pushes walls; beams sag; connections fail (WTC 2001 showed the importance of insulation & connections).

> **Concrete**: spalling (explosive flaking) due to steam pressure, loss of strength above 300 C, bond failure; use polypropylene fibres in high-strength concrete; adequate cover.

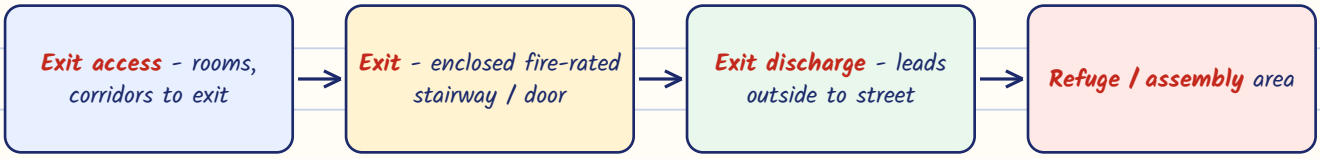
> **Redundancy and robust connections**: alternate load paths, tied structure, strong floor diaphragms, protected key elements (transfer girders, columns, stair cores).

> **Post-fire assessment**: check colour of concrete (pink above 300 C, grey above 600 C), deflection, cracks before re-occupation.

Material	Strength loss with temperature
Steel	100% at 20 C -> ~ 100% at 300 C -> ~ 50% at 550 C -> ~ 20% at 800 C
Concrete	Sound to 300 C; ~ 50% at 600 C; major loss above 800 C
Timber	Chars from outside; inner core keeps strength - predictable

VII. Concept of egress design

Egress = a continuous, unobstructed way of travel from any point in a building to a **safe place outside (public way)**. Designed so that all occupants can leave **before** conditions become untenable: **RSET < ASET** (Required Safe Egress Time less than Available Safe Egress Time).



> **RSET** = detection time + alarm / decision time + pre-movement time + travel time. **ASET** = time to reach smoke, heat, toxicity limits.

> Features: at least **two separate exits** (remote from each other), no dead-end corridors beyond about 6 m, doors open in the direction of exit, no locks that need keys.

> **Exits** enclosed in fire-resisting stair-cases (2 hr), self-closing doors, ventilation or pressurisation, direct to open air at ground.

> **Travel distance** (NBC, indicative): about 30 m for most occupancies, 45 m for industrial/storage with

sprinklers, 22.5 m for hazardous occupancy; reduce if no sprinklers.

> **Marking**: green 'EXIT' signs, floor-level photoluminescent strips, **emergency lighting** (90 min), exit signs lit by battery, plan displayed on each floor.

> **Staircase**: width, riser 190 mm max, tread 250 mm min, handrails both sides, landings; open stair and lifts are not exits.

> **Refuge areas / areas of rescue assistance** for disabled persons and for high-rise buildings; horizontal exits into the next compartment.

Egress components checklist

Component	Requirement
Doors	Min width 1.0 m, swing outward, panic bars for assembly; fire doors self-closing
Corridors	Min 1.0-1.5 m clear width (more for assembly/hospital); no obstructions
Stairs	Min 1.0-2.0 m clear width by occupancy; enclosed; 2 in number for high rise
Ramps	Slope not steeper than 1 in 12, anti-slip; for hospitals
Lifts	Not for escape in fire; except fire-lifts with special features

VIII. Exit width calculations

Exit width is calculated from the **number of occupants (occupant load)** and the **capacity of exit per unit width**. NBC uses a **unit exit width of 500 mm** (the width needed for one person stream).

Occupant load N = Floor area / Area per person ; Exit width = (N / persons per unit width) x 0.5 m

Occupancy	Floor area per person (NBC, indicative)
Residential	12.5 m2 (about)
Educational	4 m2 (classroom)
Business / offices	10 m2
Mercantile (shops)	3 m2 (ground), 6 m2 (upper)
Assembly (with fixed seats)	As per seats; unseated 0.6-1 m2
Industrial	10 m2
Storage	30 m2

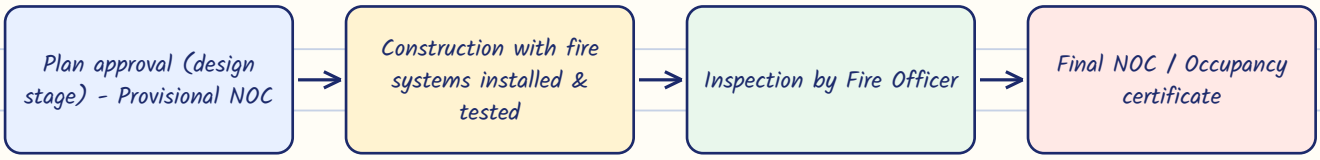
Worked example

An office floor of 600 m2. Occupant load N = 600 / 10 = **60 persons**. If **75 persons pass per unit width of door** and **50 persons per unit width of stair**: door width = (60/75) x 0.5 = **0.4 m** -> but the minimum 1.0 m is provided. Stair width = (60/50) x 0.5 = **0.6 m** -> min stair width 1.0 m governs -> provide **1.0 m, two stairs**.

> Check the **total capacity of all exits** when the largest exit is blocked (assume one exit not available).
> Width needed grows with the number of floors served: stair width at lower floors must accumulate the occupants of upper floors.
> Use of simulation (evacuation models like Pathfinder / buildingEXODUS) for large assembly and high-rise.

IX. Fire certificates (Fire NOC)

A **Fire No-Objection Certificate (NOC)** is issued by the **State Fire Service / Fire Officer** to show that the building meets the fire-safety norms. Mandatory for high-rise (15 m or more), assembly, institutional, industrial, hazardous and many other buildings; the exact thresholds are set by the State.



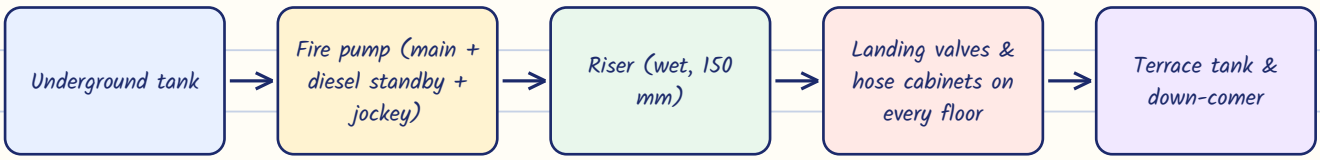
> Documents: building plans, fire-fighting layout, details of water tanks & pumps, escape routes, fire-protection design, structural-stability certificate, fire-system test reports & installer certificates, insurance.
> **Validity & renewal**: usually valid for 1-3 years (varies by State); **renewal** after re-inspection; fee; changes in use/ extension need a fresh NOC.
> **Fire insurance and licence**: occupancy/trade licence, factory licence, PESO licence often need the NOC. The Owner/Occupier is responsible for maintenance.
> Annual **fire-safety audit** by an accredited agency; mock drill logs; maintenance records.
> Non-compliance: prohibitory order, sealing, penalty, prosecution (e.g. under the State Fire Prevention and Fire Safety Acts).

X. Fire safety requirements for high-rise buildings

High-rise building: height **15 m or more** (NBC; some states use 18 m / 24 m) - fire-fighting ladders can't reach from outside, so internal systems are essential. Features follow NBC 2016 Part 4.

Requirement	Details (indicative)
Staircases	At least two (one may be external) fire-resisting enclosed stairs (2 hr); min width 1.5 m (1.0 m residential); **pressurised** (50 Pa) to keep out smoke.
Fire lift	At least 1 per 8 floors / compartment; lift lobby; capacity 545 kg (8 persons); fire-fighter's switch; speed to reach top within 1 min; standby power.
Refuge areas	Safe areas on floors for temporary shelter (about 15 m2 per 24 m of height - as per NBC) and for physically disabled.
Wet riser / down-comer	Wet riser in staircase lobbies with landing valves; 100-150 mm dia; fed by fire pumps; down-comer from terrace tank.
Fire pumps & water storage	Underground tank (about 50,000 - 2,00,000 L) + terrace tank (20,000-50,000 L); electric + diesel pump 2850 lpm; jockey pump.
Automatic sprinklers	Required in basements, high-risk areas and in buildings above 24-30 m; activated at 68 C (bulb).
Detection & alarm	Addressable smoke/heat detectors, manual call points, public address and voice evacuation, fire control room on ground floor.
Smoke management	Smoke extraction (basement 12 air changes/hr), pressurisation of stairs & lobbies, smoke curtains/shafts.
Compartmentation	Fire-rated floors, shafts, vertical ducts; fire dampers; fire stopping of services.
Emergency lighting & power	Standby DG set, exit signage, lights in stairs for 90 min.
Access for fire engines	Fire-fighting vehicle access around the building, 6 m wide with turning, 45 tonne loading; hydrants around; helipad for very tall.
Management	Fire officer, fire warden, evacuation plan, fire drills every 6 months, signage, training, annual NOC renewal.

Typical high-rise fire-fighting system layout



X-A. Supporting topics

Automatic sprinkler system

Type	Working	Use
Wet pipe	Pipes always full of water; bulb bursts at 68 C and water flows	Most buildings (simplest, most reliable)
Dry pipe	Pipes hold air under pressure; valve opens when sprinkler operates	Unheated places (freezing risk)
Pre-action	Needs detector signal + sprinkler operation	Data centres, museums
Deluge	Open sprinklers; all discharge together on detection	High hazard: aircraft hangar, transformers

> Bulb colour shows rating: orange 57 C, red 68 C, yellow 79 C, green 93 C, blue 141 C.

> Coverage about 9-12 m2 per sprinkler (light hazard); design density 2.25 - 12.5 mm/min depending on hazard class.

Smoke - the main killer

> Most fire deaths (about 60-80%) are due to **smoke and toxic gases (CO, HCN)**, not burns; smoke also blocks visibility so people lose their way.

> **Stack effect** in tall buildings drives smoke upward through shafts and stairs, so lift shafts and stairs need pressurisation or smoke lobbies.

> Smoke control: compartments, smoke curtains / screens, natural or mechanical **smoke extraction**, vents at the top, make-up air.

Fire safety management in buildings

1. Appoint a **fire safety officer** and fire wardens; prepare an emergency evacuation plan.
2. Carry out a **fire risk assessment** regularly and keep records.
3. Maintain fire systems with **inspection - testing - maintenance (ITM)** schedule: alarm weekly, extinguishers monthly, sprinklers quarterly, pumps weekly run.
4. Keep exits unlocked and clear; no storage in stair-cases; check fire doors are not wedged open.
5. Train staff and conduct fire drills; give induction to new occupants and visitors.
6. Control hot work, electrical load, waste, and decorative materials; use fire-retardant furnishings.

Lessons from major building fires

Fire	What went wrong
Triangle Shirtwaist, NY (1911)	Locked exits, blocked stairs; 146 deaths - start of fire code.
Uphaar Cinema, Delhi (1997)	Transformer fire, smoke in balcony, blocked exits and extra seats; 59 died.
AMRI Hospital, Kolkata (2011)	Basement stored flammables; smoke spread, locked windows, no alarm; 90+ died.
Kamala Mills, Mumbai (2017)	Illegal rooftop restaurants, flammable decor, no NOC; 14 died.
Grenfell Tower, London (2017)	Combustible cladding, single stair, no sprinklers; 72 died.
Takshashila Arcade, Surat (2019)	Fire at coaching centre with no exits; students jumped; 22 died.

Safe building = **Compartment** + **Escape** + **Detection** + **Suppression** + **Management**

Egress example 2 - assembly hall

A hall for 600 persons: exits for assembly at 100 persons per unit width of door (0.5 m): width = (600/100) x 0.5 = **3.0 m total**. Provide **3 exits of 1.5 m** (so that loss of one still leaves 3.0 m). Stairs at 50 persons per unit: (600/50) x 0.5 = 6 m of stairs.

XI-A. Active & passive systems in buildings - summary

Passive (built-in)	Active (operated)
Fire-rated walls, floors, shafts	Detectors, alarm, voice evacuation
Fire doors, dampers, fire stops	Sprinklers, deluge, gas / foam systems
Protected stairs, corridors, refuge	Hydrants, wet riser, hose reels, extinguishers
Structural fire protection, cladding choice	Smoke extraction, stair pressurisation, fire lift
Separation distance, fire-resistant roofs	Emergency lighting & standby generator

Fire safety signs (IS 12349 / ISO 7010)

Sign colour / shape	Meaning	Example
Green square	Escape route, exit, first aid	Running man + arrow
Red square	Fire-fighting equipment	Extinguisher, hydrant, call point

Sign colour / shape	Meaning	Example
Red circle with bar	Prohibition	No smoking, no naked flames, do not use lift in fire
Blue circle	Mandatory	Keep fire door closed

Types of construction (NBC) in short

> **Type 1**: all structural parts non-combustible, fire resistance 4 hr (columns), 3 hr (beams) - high-rise, hospitals.

> **Type 2**: 3 hr columns, 2 hr beams. **Type 3**: 2 hr columns, 2 hr beams. **Type 4**: 1 hr, mainly timber / low-rise.

> Choice depends on **height, floor area and occupancy hazard**; large buildings with higher fire load demand higher type.

Quick formulas of the unit

Formula	Use
q = SUM(M x H) / A	Fire load density (MJ/m2)
T = 345 log10 (8t + 1) + 20	ISO 834 standard fire temperature
N = Area / Area per person	Occupant load
W = (N / persons per unit) x 0.5 m	Required exit width
RSET < ASET	Safe egress condition

XI. Quick revision & important questions

Remember

> Fire load = sum(M x H) / A MJ/m2 (low / moderate / high).

> Fire resistance: stability, integrity, insulation; ISO 834 curve; steel critical about 550 C.

> Egress: exit access, exit, exit discharge; RSET < ASET; at least 2 exits.

> Unit exit width 500 mm; occupant load = area / area per person.

> High-rise = 15 m +; 2 stairs, fire lift, wet riser, sprinkler, refuge, fire NOC.

1. State the objectives of fire-safe building design. (5 M)
2. Define fire load. How is it calculated? Classify occupancies by fire load. (10 M)

3. Explain fire-resistant materials and fire resistance testing. (10 M)
4. Describe methods of structural fire protection and structural integrity. (10 M)
5. Explain the concept of egress design and calculate the exit width for a given building. (10 M)
6. What are fire certificates? Procedure to obtain Fire NOC. (5 M)
7. Discuss fire safety requirements for high-rise buildings. (10 M)